

6 The Gamma and Zeta Functions

It is no exaggeration to say that the gamma and zeta functions are among the most important nonelementary functions in mathematics. The gamma function Γ is ubiquitous in nature. It arises in a host of calculations and is featured in a large number of identities that occur in analysis. Part of the explanation for this probably lies in the basic structural property of the gamma function, which essentially characterizes it: $1/\Gamma(s)$ is the (simplest) entire function¹ which has zeros at exactly $s = 0, -1, -2, \dots$

The zeta function ζ (whose study, like that of the gamma function, was initiated by Euler) plays a fundamental role in the analytic theory of numbers. Its intimate connection with prime numbers comes about via the identity for $\zeta(s)$:

$$\prod_p \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

where the product is over all primes. The behavior of $\zeta(s)$ for real $s > 1$, with s tending to 1, was used by Euler to prove that $\sum_p 1/p$ diverges, and a similar reasoning for L -functions is at the starting point of the proof of Dirichlet's theorem on primes in arithmetic progression, as we saw in Book I.

While there is no difficulty in seeing that $\zeta(s)$ is well-defined (and analytic) when $\operatorname{Re}(s) > 1$, it was Riemann who realized that the further study of primes was bound up with the analytic (in fact, meromorphic) continuation of ζ into the rest of the complex plane. Beyond this, we also consider its remarkable functional equation, which reveals a symmetry about the line $\operatorname{Re}(s) = 1/2$, and whose proof is based on a corresponding identity for the theta function. We also make a more detailed study of the growth of $\zeta(s)$ near the line $\operatorname{Re}(s) = 1$, which will be required in the proof of the prime number theorem given in the next chapter.

¹In keeping with the standard notation of the subject, we denote by s (instead of z) the argument of the functions Γ and ζ .

1 The gamma function

For $s > 0$, the **gamma function** is defined by

$$(1) \quad \Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt.$$

The integral converges for each positive s because near $t = 0$ the function t^{s-1} is integrable, and for t large the convergence is guaranteed by the exponential decay of the integrand. These observations allow us to extend the domain of definition of Γ as follows.

Proposition 1.1 *The gamma function extends to an analytic function in the half-plane $\operatorname{Re}(s) > 0$, and is still given there by the integral formula (1).*

Proof. It suffices to show that the integral defines a holomorphic function in every strip

$$S_{\delta, M} = \{\delta < \operatorname{Re}(s) < M\},$$

where $0 < \delta < M < \infty$. Note that if σ denotes the real part of s , then $|e^{-t} t^{s-1}| = e^{-t} t^{\sigma-1}$, so that the integral

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt,$$

which is defined by the limit $\lim_{\epsilon \rightarrow 0} \int_\epsilon^{1/\epsilon} e^{-t} t^{s-1} dt$, converges for each $s \in S_{\delta, M}$. For $\epsilon > 0$, let

$$F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t} t^{s-1} dt.$$

By Theorem 5.4 in Chapter 2, the function F_ϵ is holomorphic in the strip $S_{\delta, M}$. By Theorem 5.2, also of Chapter 2, it suffices to show that F_ϵ converges uniformly to Γ on the strip $S_{\delta, M}$. To see this, we first observe that

$$|\Gamma(s) - F_\epsilon(s)| \leq \int_0^\epsilon e^{-t} t^{\sigma-1} dt + \int_{1/\epsilon}^\infty e^{-t} t^{\sigma-1} dt.$$

The first integral converges uniformly to 0, as ϵ tends to 0 since it can be easily estimated by ϵ^δ/δ whenever $0 < \epsilon < 1$. The second integral converges uniformly to 0 as well, since

$$\left| \int_{1/\epsilon}^\infty e^{-t} t^{\sigma-1} dt \right| \leq \int_{1/\epsilon}^\infty e^{-t} t^{M-1} dt \leq C \int_{1/\epsilon}^\infty e^{-t/2} dt \rightarrow 0,$$

and the proof is complete.

1.1 Analytic continuation

Despite the fact that the integral defining Γ is not absolutely convergent for other values of s , we can go further and prove that there exists a meromorphic function defined on all of $\mathbb{C}$ that equals Γ in the half-plane $\operatorname{Re}(s) > 0$. In the same sense as in Chapter 2, we say that this function is the analytic continuation² of Γ , and we therefore continue to denote it by Γ .

To prove the asserted analytic extension to a meromorphic function, we need a lemma, which incidentally exhibits an important property of Γ .

Lemma 1.2 *If $\operatorname{Re}(s) > 0$, then*

$$(2) \quad \Gamma(s+1) = s\Gamma(s).$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \dots$

Proof. Integrating by parts in the finite integrals gives

$$\int_{\epsilon}^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s) dt = - \int_{\epsilon}^{1/\epsilon} e^{-t}t^s dt + s \int_{\epsilon}^{1/\epsilon} e^{-t}t^{s-1} dt,$$

and the desired formula (2) follows by letting ϵ tend to 0, and noting that the left-hand side vanishes because $e^{-t}t^s \rightarrow 0$ as t tends to 0 or ∞ . Now it suffices to check that

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1,$$

and to apply (2) successively to find that $\Gamma(n+1) = n!$.

Formula (2) in the lemma is all we need to give a proof of the following theorem.

Theorem 1.3 *The function $\Gamma(s)$ initially defined for $\operatorname{Re}(s) > 0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the negative integers $s = 0, -1, \dots$. The residue of Γ at $s = -n$ is $(-1)^n/n!$.*

²Uniqueness of the analytic continuation is guaranteed since the complement of the poles of a meromorphic function forms a connected set.

Proof. It suffices to extend Γ to each half-plane $\operatorname{Re}(s) > -m$, where $m \geq 1$ is an integer. For $\operatorname{Re}(s) > -1$, we define

$$F_1(s) = \frac{\Gamma(s+1)}{s}.$$

Since $\Gamma(s+1)$ is holomorphic in $\operatorname{Re}(s) > -1$, we see that F_1 is meromorphic in that half-plane, with the only possible singularity a simple pole at $s = 0$. The fact that $\Gamma(1) = 1$ shows that F_1 does in fact have a simple pole at $s = 0$ with residue 1. Moreover, if $\operatorname{Re}(s) > 0$, then

$$F_1(s) = \frac{\Gamma(s+1)}{s} = \Gamma(s)$$

by the previous lemma. So F_1 extends Γ to a meromorphic function on the half-plane $\operatorname{Re}(s) > -1$. We can now continue in this fashion by defining a meromorphic F_m for $\operatorname{Re}(s) > -m$ that agrees with Γ on $\operatorname{Re}(s) > 0$. For $\operatorname{Re}(s) > -m$, where m is an integer ≥ 1 , define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

The function F_m is meromorphic in $\operatorname{Re}(s) > -m$ and has simple poles at $s = 0, -1, -2, \dots, -m+1$ with residues

$$\begin{aligned} \operatorname{res}_{s=-n} F_m(s) &= \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} \\ &= \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} \\ &= \frac{(-1)^n}{n!}. \end{aligned}$$

Successive applications of the lemma show that $F_m(s) = \Gamma(s)$ for $\operatorname{Re}(s) > 0$. By uniqueness, this also means that $F_m = F_k$ for $1 \leq k \leq m$ on the domain of definition of F_k . Therefore, we have obtained the desired continuation of Γ .

Remark. We have already proved that $\Gamma(s+1) = s\Gamma(s)$ whenever $\operatorname{Re}(s) > 0$. In fact, by analytic continuation, this formula remains true whenever $s \neq 0, -1, -2, \dots$, that is, whenever s is not a pole of Γ . This is because both sides of the formula are holomorphic in the complement of the poles of Γ and are equal when $\operatorname{Re}(s) > 0$. Actually, one can go further, and note that if s is a negative integer $s = -n$ with $n \geq 1$, then both sides of the formula are infinite and moreover

$$\operatorname{res}_{s=-n} \Gamma(s+1) = -n \operatorname{res}_{s=-n} \Gamma(s).$$

Finally, note that when $s = 0$ we have $\Gamma(1) = \lim_{s \rightarrow 0} s\Gamma(s)$.

An alternate proof of Theorem 1.3, which is interesting in its own right and whose ideas recur later, is obtained by splitting the integral for $\Gamma(s)$ defined on $\operatorname{Re}(s) > 0$ as follows:

$$\Gamma(s) = \int_0^1 e^{-t} t^{s-1} dt + \int_1^\infty e^{-t} t^{s-1} dt.$$

The integral on the far right defines an entire function; also expanding e^{-t} in a power series and integrating term by term gives

$$\int_0^1 e^{-t} t^{s-1} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)}.$$

Therefore

$$(3) \quad \Gamma(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t} t^{s-1} dt \quad \text{for } \operatorname{Re}(s) > 0.$$

Finally, the series defines a meromorphic function on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $s = -n$. To prove this, we argue as follows. For a fixed $R > 0$ we may split the sum in two parts

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} = \sum_{n=0}^N \frac{(-1)^n}{n!(n+s)} + \sum_{n=N+1}^{\infty} \frac{(-1)^n}{n!(n+s)},$$

where N is an integer chosen so that $N > 2R$. The first sum, which is finite, defines a meromorphic function in the disc $|s| < R$ with poles at the desired points and the correct residues. The second sum converges uniformly in that disc, hence defines a holomorphic function there, since $n > N > 2R$ and $|n+s| \geq R$ imply

$$\left| \frac{(-1)^n}{n!(n+s)} \right| \leq \frac{1}{n!R}.$$

Since R was arbitrary, we conclude that the series in (3) has the desired properties.

In particular, the relation (3) now holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

The following identity reveals the symmetry of Γ about the line $\operatorname{Re}(s) = 1/2$.